CIVITAS
Sustainable and smart mobility for all
2030

CIVITAS indicators

Average distance per delivery (TRA_FR_EF1)

DOMAIN

Transport	Environment	Energy	Society	Economy

TOPIC	**Freight**	
IMPACT	**Operational efficiency of urban deliveries** *Improved efficiency of freight distribution through route optimisation and other sustainable urban freight interventions.*	**TRA_FR**

Category

Key indicator	Supplementary indicator	State indicator

CONTEXT AND RELEVANCE

Urban freight measures frequently aim to reduce the distance travelled for each delivery while maintaining service quality. A reduction in average distance per delivery indicates more efficient logistics organisation, improved vehicle utilisation, better spatial allocation of demand, or increased consolidation of deliveries. These improvements can help reduce unnecessary vehicle kilometres in urban areas and contribute to lower congestion, emissions, noise, road safety risks, and pressure on scarce public space.

This indicator measures the average distance travelled by freight delivery vehicles per completed delivery within the pilot area. **It is a relevant indicator when the policy action aims to increase the efficiency of urban freight distribution. A successful action is reflected in a LOWER value of the indicator after the experiment compared to the BAU case.**

DESCRIPTION

The indicator assesses how efficiently delivery routes are organised and whether freight interventions reduce unnecessary travel per delivery.

The indicator is expressed as a ratio of **km/delivery**.

METHOD OF CALCULATION AND INPUTS

The indicator should be calculated **exogenously** based on the required inputs and then coded in the supporting tool.

Method 1

Survey delivery companies on a sample of delivery tours	Significance: **0.50**

INPUTS

The following information is needed to compute the indicator:

- The **total number of deliveries** made in the pilot area by the logistics operators surveyed over a defined reference period, including only motorised modes.
- The **total vehicle-kilometres travelled** in the pilot area and over the reference period by the motorised freight vehicles used to complete the recorded deliveries.

EQUATIONS

The equation that should be applied to quantify the indicator is:

$$\boldsymbol{AvgDistDel} = \frac{\sum_i Distance_i}{\sum_i MDeliveries_i}$$

Where:

$Distance_i$= Distance travelled by delivery vehicle i during the observation period (km)

$MDeliveries_i$= Number of deliveries completed by motorized vehicle i during the observation period

ALTERNATIVE INDICATORS

This indicator measures the average distance travelled per delivery. Other indicators in the CIVITAS Evaluation Framework addressing the operational efficiency of urban deliveries include: **TRA_FR_EF2**, which measures the share of unsuccessful deliveries; **TRA_FR_EF3**, which measures the average time per delivery; and **TRA_FR_EF4** and **TRA_FR_EF5**, which measure the average load factor of delivery vehicles for B2B and B2C deliveries, respectively.